A Survey Report on Recent Progresses in Nearest Neighbor Realization of Quantum Circuits

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Abstract. The thrive of attending very fast computation has breeds the ideas for new computing paradigms like optical computing, quantum computing and in last few years, quantum computing has left an impressive footprint before the design industry. Even companies like IBM, Microsoft are claiming the initiation of physical implementation of quantum circuits in on-chip units. But the successful implementation of this circuit needs to satisfy several design constraints and one such constraint is Nearest Neighbor (NN) enforcement.
To introduce the reader with the progresses made in NN-based implementation of quantum circuits, here, in this survey paper we have tried to include some of the peer-reviewed works in our content. While describing the works, we have added necessary examples so that it becomes easy to follow for a reader. To this extent, we also have introduced the NN logic and its related cost functions in our survey report.

Keywords: Quantum Circuit, quantum gate, Nearest Neighbour (NN), SWAP gate , Quantum Cost(QC).

1 Introduction

Emergence of quantum computing has made an alluring effect among the research community due to its proficiency in solving some complicated problems with exponential speedup over conventional computing paradigm. As a result, realization towards a highly scalable and fault tolerant quantum architecture [1] received tremendous attention. However, there exist many difficulties that prevent the practical realization of a real time quantum computing device. In this conjecture, fault-tolerance behavior is considered as the essential driving factor towards accomplishments of quantum circuit design. For this purpose, error-correcting codes like surface or steane codes found significant [1]. Moreover, these codes become applicable only for nearest neighbor quantum architecture. In other words, implementation of error correcting codes demands the qubits to interact with only adjacent neighbors. In addition to this, such nearest neighbor qubit interaction has also been regarded as the essential design

parameter for the physical realization of some quantum technologies like ion-trap [2], quantum dots [3], superconducting qubits [4] and nuclear magnetic resonance [5]. Consequently, ensuring of nearest neighbor architecture is considered as the necessary design issue for quantum circuit synthesis.

Establishment of such nearest neighbor architecture can be acquired by making the non-adjacently placed qubits adjacent with the help SWAP gates. It can be carried out by exchanging the states of the qubits till they become adjacent. Following such a SWAP insertion procedure in turn enhances the design overhead of the resultant NN circuit. Optimization of the resultant NN architecture in terms of limited use of SWAP gates has become more challenging for the researchers such that the circuit overhead can be kept under control. In this regard, several research articles related to the efficient realization of NN structures has been discussed whereby the authors considered SWAP optimization issue using various design algorithms. Here, in this survey paper we are reviewing the progress made towards efficient realization of Nearest Neighbour quantum circuit.

The rest of the paper is organized as follows. In section II, we have introduced the preliminaries on quantum circuit and the nearest neighbor constraint. A survey report on existing works of NN based design is presented in Section III and finally, we conclude the paper in section IV.

2 Background

Qubit is considered as the basic quantum information unit which can be modified using a sequence of quantum gates. This quantum unit is similar to bits used in classical computing but unlike bit, qubit has numerous states in-between 0 and 1 – known as superposition state. This superposition state can be expressed using the following expression:

$$|\Psi\rangle = \alpha|0\rangle + \beta|1\rangle \tag{1}$$

where the terms α and β are the complex probability amplitudes of the respective basis states $|0\rangle$ and $|1\rangle$ provided the condition $\alpha^2 + \beta^2 = 1$ is satisfied.

Basically, quantum gates perform quantum operations on the states of the qubit which can be realized by means of unitary matrices. In other words, quantum computations on an n-qubit system are carried out by multiplying relevant $2^n \times 2^n$ unitary matrices.

Definition 1: *Quantum gates are considered as the basic quantum operators used for manipulating qubits and a cascade of such gates along with the set of circuit lines on which they acts together forms a quantum circuit.*

Quantum gates employed in quantum circuit realization belongs to universal quantum gate libraries. NCV [6-7] is considered as the most widely used gate library containing basic quantum gates like NOT, CNOT, controlled-V and controlled-$V^\dagger$ respectively (the symbolic notation of basic gates has been depicted in Table 1).

For implementing quantum circuit, nearest neighbor qubit interaction condition needs to be preserved. In order to arrive towards such a design constraint, SWAP gates (symbolically represented in Fig. 1) are inserted in front of gates with non-

adjacent interacting qubits and interchange their locations till they become adjacent. This procedure of making the interacting qubits adjacent is considered as the Nearest Neighbor (NN) phenomenon.

Definition 2: *Nearest Neighbor cost (NNC) is considered as the distance separating the positions of the input qubits (control/target) of any 2-qubit gate and such difference in distance value can be computed as*

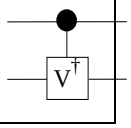
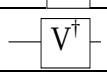
$$NNC\ (g) = |c - t| - 1 \tag{2}$$

The expression shown above estimates the nearest neighbor cost of any specific gate *g* with its control and target qubits placed at locations *c* and *t* respectively. Combining such costs of the individual gates leads to an overall cost (NNC (C)) for the given quantum circuit C and can be expressed as

$$NNC\ (C) = \sum NNC\ (g) \tag{3}$$

By investigating the above expression it can be inferred that a given circuit satisfies the nearest neighbor property if it either contains 1-qubit gates or having only 2-qubit gates with qubits placed at adjacent positions. More precisely, if the cost estimation expression given in (3) evaluates to zero then the circuit can be considered as NN-compliant.

Table 1: Symbolic representation of some quantum gates

Gates	Symbol	Gates	Symbol
NOT		Controlled -V	
CNOT			
V		Controlled - $V^\dagger$	
$V^\dagger$			

Example 1: *Consider a toffoli gate shown in Fig. 2(a) and its equivalent NCV representation. By investigating Fig. 2(b), it can be inferred that this NCV structure does not satisfy the nearest neighbor criteria as the NNC of the circuit turns out to be more than zero (NNC(C)=1). To transform this circuit into its equivalent NN realization SWAP gates are implemented and the resultant circuit is depicted in Fig. 2(c).*

Fig. 2(a): Toffoli gate Structure

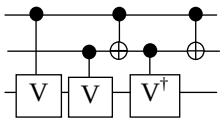


Fig. 2(b): NCV realization of Fig. 2(a)

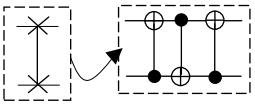


Fig.1:SWAP gate structure

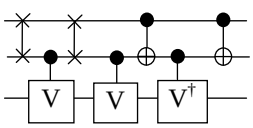


Fig. 2(c): NN structure of Fig. 2(b)

3 Works on NN Based Representation

Last couple of years has witnessed several progresses in the field of NN based representation, where the authors have mainly worked towards efficient realization of such designs. Not only embedding SWAP gates in designs remains the sole objective of these works, but incorporation of several design strategies like global reordering, local reordering, windowing policy have been followed to contain the usage of SWAP gates. On the other side, the representation of NN circuit has changed from 1D to 2D and even the concept of n-D has come into picture. Here, we have tried to include some of the peer-reviewed works and their contributions.

To achieve circuits with linear NN architecture, the existing design workflows has been extended in the article [8]. Here, template matching based post optimization approach, exact design methods as well as reordering strategies viz. Global and Local are introduced. The existing synthesis algorithms may not generate an optimal circuit with respect to the number of elementary gates used to realize the desired function so the quantum cost of the resultant circuit is not minimal. To simplify the resultant circuits, template based optimization strategies are applied. Furthermore, exploitation of such template approaches not only reduces the quantum cost but also realizes optimal NN–compliant circuit thereby several NNC based templates are designed. The idea behind using templates is to replace a set of gates by another set while at the same time preserving the functionality of the circuit. After implementing templates, the circuit is further NN optimized using traditional matching rules like deletion and moving.

For better interpretation, a NN circuit shown in Fig. 3(a) is considered and the equivalent NN optimal circuit obtained after using template matching rules is represented in Fig. 3(b).

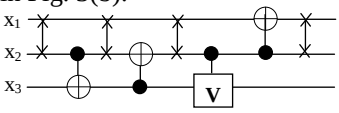


Fig. 3(a): NN circuit having quantum cost of 16

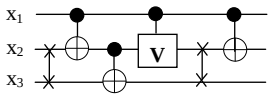


Fig. 3(b): NN circuit with quantum cost of 10

Besides using templates, an optimal circuit can also be generated by employing exact design methodology. The purpose of using such exact synthesis approaches is to construct an optimal quantum architecture both in terms of minimal quantum cost as well as NN optimality. To this end, the design problem has been expressed in the form of a SAT solving instance as discussed in article [8].

Example 2: *Consider again the Toffoli gate represented in Fig. 2(a) and also its corresponding decomposed network as depicted in Fig. 2(b). The non-optimal circuit in Fig. 2(b) is optimized with respect to NNC using the exact design approach of [9] is shown in Fig. 4(d) over the one obtained using naïve approach (see Fig. 4(c)).*

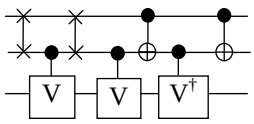


Fig. 4(c): NN structure using naïve method (quantum cost is 11)

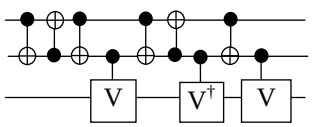


Fig 4(d): Improved NN circuit with quantum cost of 9

Although, an efficient NN quantum architecture is obtained using the previous design strategies but these solutions can be improved further by changing only the initial ordering of the qubits. Qubit reordering techniques reduces some additional NN overhead in the circuit thereby produces a better NN structures. To this end, a couple of heuristic design schemes have been employed in [8] of which one of them rearranges the qubits from a global perspective while the other one changes the line order locally.

To clearly understand the heuristics used for determining an appropriate qubit ordering is illustrated in the example presented next.

Example 3: *Considering the circuit shown in Fig. 5(a) whose original qubit ordering is changed globally. The contribution of each qubit towards NNC in the circuit is computed and based on this estimated value the qubits are reordered accordingly. The resultant reordered circuit after global reordering is depicted in Fig. 5(b). However, application of local reordering strategy on the circuit (see Fig. 5(a) produces an equivalent NN-complaint circuit in Fig. 5(c).*

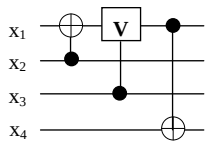


Fig. 5(a): Sample Circuit

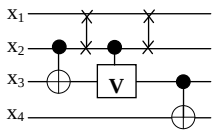


Fig. 5(b): Globally reordered circuit of Fig. 5(a)

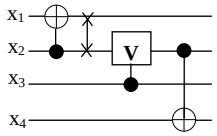


Fig. 5(c): locally reordered circuit of Fig. 5(a)

All the design workflows discussed so far are integrated together and appended to the existing quantum circuit synthesis algorithms so as to explicitly consider the NN restriction into account and such an attempt has been the underlying proposition of the work [8].

The NN design structures obtained using the approaches stated in [8] are not optimal so there exists opportunity for further improvement. For this purpose, NN optimization of quantum circuits in single dimensional representation has been addressed in [9]. Here, nearest neighbor optimization problem has been transformed into a Minimum Linear Arrangement (MINLA) problem. Initially the optimization problem is represented in the form of an interaction graph for a given circuit. Thereafter, the given circuit is partitioned into sub-circuits where qubits are reordered for each sub-circuits such that a localized circuit with adjacent gates is obtained. For this purpose, SWAP gates are implemented in between sub-circuits so as to generate the desired qubit ordering for each of them. This has been done by constructing interaction graphs for each sub-circuit and then MINLA approach is applied on them separately. In other words, each non-adjacent gate in each sub-circuit needs to be made NN-compliant by locally rearranging the qubits order involved in the sub-circuits. The solutions obtained after solving each sub-circuits may not generate a unique result and to resolve this the one that minimizes the total SWAP gate requirements in between all the sub-circuits is selected otherwise anyone is chosen randomly. Additionally, a look-ahead strategy is implemented to improve the solutions obtained using MINLA.

For better realization of the proposed approach, a sample benchmark circuit named 4gt11_84 has been considered as shown in Fig. 6(a). From Fig. 6(a), it can be discerned that the given circuit contains non-local gates (non-adjacent) so to trans-

form it into a local circuit the proposed MINLA approach is implemented and the resultant circuit is depicted in Fig. 6(b).

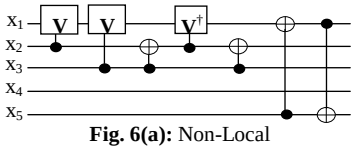


Fig. 6(a): Non-Local benchmark circuit (4gt11_84)

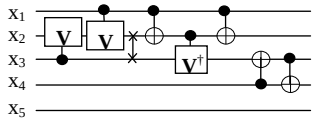


Fig. 6(b): Local circuit of Fig. 6(a) obtained using MINLA

All these existing linear NN design methodologies are based on heuristic policy which produces sub-optimal or near optimal solutions. An alternative synthesis workflow involving exact methodology based on constraint solvers has been introduced in [10]. Here, both global and local reordering strategies are discussed determining the best linear qubit order sequence thereby producing an optimal NN circuit.

To find the best global qubit order, the NNC value is recomputed after generating every permutation and comparing with the one obtained in previous sequence rather than considering the NNC impact of each qubit as done in [8]. For this purpose, initially an adjacency matrix of a given circuit is constructed and then a modified cost function presented in [10] is employed for evaluating such qubit sequences.

To have a better interpretation, an example is provided illustrating the realization of globally ordered quantum circuit using the exact NN optimization scheme stated in [10].

Example 4: *Consider the quantum circuit represented in Fig. 7(a) with initial qubit order $\pi(q_0q_1q_2q_3q_4)$. Execution of an exact reordering strategy stated in [10] obtains an optimal NN based representation requiring only two SWAP gates (see Fig. 7(b) compared to the architecture achieved using heuristic method (see Fig. 7(c)) stated in [8].*

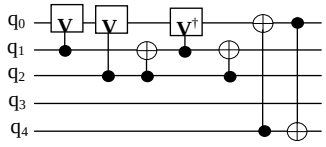


Fig. 7(a): Input quantum circuit

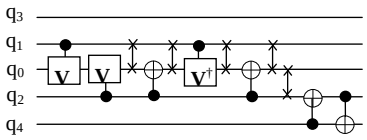


Fig. 7(b): NN circuit obtained using exact global optimization approach of [10]

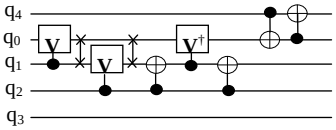


Fig. 7(c): NN circuit obtained using heuristic global optimization approach [8]

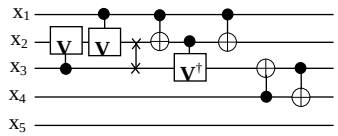


Fig. 8(c): NN representation using look-ahead only on the next immediate gate or on one gate

Like global approach, an exact local optimization method based on pseudo Boolean optimization solvers is discussed in [10] for producing optimal NN solutions. Determining the best local qubit ordering sequence from all possible options can be done by evaluating the cost using inversion vectors. The resultant value obtained after combining the individual entries of inversion vector corresponds to the amount of SWAP operations needed to create a specific qubit sequence order. However, such

estimation procedure does not turn out to be beneficial for qubit rearrangement. Therefore, possible permutation options for all gates in the given circuit are considered and evaluated using cost function stated in [10]. Finally, the ordering having a least cost value is implemented. To find such best order within reasonable time bound, the design problem has been formulated in the form of Boolean satisfiability representation as discussed in [10].

Although implementation of exact design methodology produces an optimal NN-complaint circuit but such strategies hardly found feasible for circuits with large number of qubits due to huge computational time requirements. For this purpose, an efficient heuristic synthesis workflow based on local reordering approach has been presented in the article [11] that improves the NN architecture in terms of cost reduction and also satisfies scalability. For such efficient NN realization, an *N*-gate based look-ahead methodology has been employed which makes any non-adjacent gate *g* adjacent by observing the effect on *N* following gates and inserts SWAP gates in such a manner so that less SWAP gate are required for the following gates.

However, there can be instances in which change of either control/target qubit positions of any non-adjacent gate may not have any effect on the following gates. All such possible cases are identified and categorized into four different classes based on various gate combinations. These classes are formed estimating the effect of moving the control/target qubit positions of any current gate g_c by considering a pair of each *N* gates. For each of these classes, control and target qubit positions are permuted in all possible manners so as to obtain the total SWAP count needed to move the control or target qubit of g_c downwards. In this manner a given circuit is made NN-compliant by scanning it from the left and inserting SWAP gates as needed.

To get a better apprehension of the SWAP cost calculation an illustrative example has been provided next.

Example 5: *Consider a benchmark function shown in Fig. 8(a) which needs to be made NN-complaint using heuristic look-ahead scheme stated in [11]. The equivalent NN design obtained using look-ahead process by considering only three following gates is represented in Fig. 8(b). An improved solution with less NNC can be realized when applying look-ahead process considering only a single gate is depicted in Fig. 8(c).*

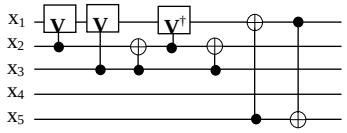


Fig. 8(a): Benchmark Function (4gt11_84)

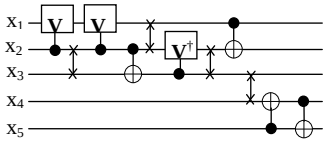


Fig. 8(b): NN representation using look-ahead on three gates

All these design approaches transforms a given quantum circuit to its equivalent linear NN representations. But such architecture restricts the qubits to have only two adjacent neighbors which in turn increases the design overhead in the resultant NN circuit. However, technological advancements lead to the development of 2D representation whereby the number of interacting neighbors rises from two to four for a qubit. For this purpose, the authors of [12] presented a synthesis automation model

that improves the qubit interactions for quantum technologies using 2D representation. To reflect such higher dimensional NN design, some placement procedure needs to be developed for arranging the qubits in 2D structures and an attempt based on Mixed Integer Programming (MIP) is employed in the work [12]. Here, the synthesis workflow uses two methods of which one of them (Method 1) is relied only on single grid based MIP model where SWAP gates are simply added before any non-adjacent gates after initial qubit arrangement. In the other method (Method 2), the mapping problem is solved by partitioning the given circuit into sub-circuits and for each of them qubit mappings is determined and SWAP gates are added in each sub-circuit similar to Method 1. In addition to this, a swapping network is needed to transform the qubit placement derived in one sub-circuit to the placement used in the other one. To realize this swapping between any two consecutive sub-circuits, a bubble sort algorithm is implemented on the 2D structure. The 2D results obtained from the proposed design methods in [12] (Method 1, Method 2) clearly outperforms the best 1D results by reducing the SWAP gate count substantially.

The 2D result obtained in [12] is improved further by employing fast and efficient qubit mapping scheme which has been the underlying objective of the work [13]. The significance of appropriate grid size along with initial qubit mapping has been discussed in [13]. In this regard, finding proper grid size is considered as the essential optimization problem and addressed by formulating heuristic estimation models. Following this, the input qubits are arranged on such 2D grid by running an initial qubit mapping algorithm. For this purpose, a priority stack is used defining the order of selecting the qubits and subsequently placed on the grid structure using the corresponding qubits interactions. Finally, SWAP gate insertion process is followed whereby row and column difference of the non-adjacent interacting qubits is computed such that they can be made adjacent by moving them either horizontally or vertically.

The NN results obtained in [13] can be improved further by exploiting even better qubit mapping approach and this has been addressed in the works [14, 15] whereby the entire process is segmented into three stages of qubit selection, qubit placement and SWAP insertion. Selection and placement of qubits are conducted based on some metrics developed using the interaction details of the qubits in the given circuits. In [14], qubits are arranged onto a 2D structure using interaction time of the corresponding qubits while cost factor metric is estimated for qubit mapping purpose discussed in [15]. Finally, SWAP gates are inserted by examining each gate in the given circuit leads to an NN-compliant circuit.

The workflow introduced in [16] discusses an efficient heuristic mapping approach providing better results compared to previous approaches irrespective of any specific dimensional representations. Here, a global reordering scheme is designed using both exhaustive and genetic algorithm based search procedure. Afterwards, a local reordering process considering all possible paths occurring between any two interacting qubits is implemented. Both the reordering schemes employs an improved cost estimation function stated in [16]. Though use of exhaustive strategy produces optimal results but found infeasible for large circuits and thereby an alternative heuristic based genetic algorithm is applied. After populating the qubits on given architecture, SWAP gates can be inserted in various ways considering only the one having least cost value

on a path chosen randomly from the available paths. Furthermore, the proposed method can be extended to higher or multi-dimensional architectures.

Example 7: *Consider the circuit shown in Fig. 9 where the dashed region marked on the left side of the circuit designates the set of gates with adjacent interacting qubits after performing SWAP operations while the one on the right represents the gates with non-adjacent qubits. To make these gates adjacent, SWAP gates need to be applied in the best possible manner for a given path. The arrows shown in Fig. 9 indicates that there exists two possible ways in which SWAP gate can be inserted to make the corresponding gate adjacent.*

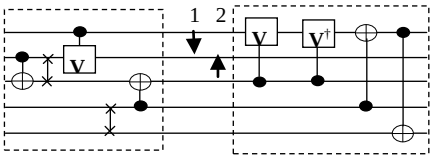


Fig. 9: SWAP insertion process

4 Conclusion

NN constraint is found to be one of the major hindrances towards physical implementation of quantum circuit and to highlight the progresses made in this field, in this paper, we have surveyed some of the state-of-art design techniques and their contributions. Starting from 1D representation to 2D and in later part we have surveyed the works on n-D NN architecture as well. Strategies like gate reordering, line reordering, windowing techniques are also included in this review paper.

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